

# Pushkar Dangwal

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**Software Development Engineer** and final-year Computer Science undergraduate at Sharda University, with hands-on experience in **full-stack development, secure systems, and AI/ML integration**. Published twice in **IEEE on security frameworks and self-sovereign identity**, with a LeetCode rank under 70,000 (**700+ problems solved**). Contributed as an **SDE Intern at Delhi Police**, building full-stack operational tools deployed in live environments. Core strengths span **ReactJS, Node.js, Python, Kotlin, and distributed system design**, with a working approach grounded in writing secure, maintainable code for problems that demand precision.

## EDUCATION

**BACHELOR OF TECHNOLOGY COMPUTER SCIENCE**, Sharda University  2022 – Present  
CGPA: 8.72 | 1st prize, Technokrats 2023 | LeetCode rank under 70,000 (700+ problems) Greater Noida, India

**XII Class**, Ganga International School  2021 – 2022  
Marks : 94.33% Hiran Kudna, New Delhi

## WORK HISTORY

**Software Development Engineer Intern**, Delhi Police 12/2024 – 02/2025  
Rohini, New Delhi

- Built operational tools for law enforcement use including a **Kotlin Android app**, a **WebRTC based communication system** and a **Next.js Firebase** real-time dashboard.
- Integrated **Firebase Realtime Database** with a **native Android frontend** to enable field officers to retrieve case-critical information on the go, **reducing lookup time from several minutes to under 30 seconds**.
- Applied **secure coding standards** across all deliverables including **encrypted data transmission**.

## PUBLICATIONS

**Towards Trustworthy Digital Identity: Designing a Privacy-Enhancing Self-Sovereign Identity Framework**, 03/2026  
Presented at IC-ICNS (IEEE) 2026 C. V. Raman Global University.

**Prescriptive Security Framework for Managing risks and Quantifying Security in Software Development Practices** , IEEE 12/2024

## PROJECTS

**Blockchain-Based Identity Verification and Fraud Detection System**  08/2025 – 04/2026

- Built a decentralised identity verification system using **Solidity smart contracts** for tamper-proof credential management, directly backed by the **IEEE 2026** paper on self-sovereign identity.
- Implemented **credential issuance, verification and revocation flows** end to end with a React based frontend for secure on-chain identity management.
- Designed a separate ML fraud detection pipeline using **Deep Isolation Forest and LightGBM** for real-time anomaly detection and transaction risk scoring.

**Technology/Tools:** Blockchain (Smart Contracts), ReactJS, Node.js, Python, LightGBM, Isolation Forest, Firebase

**CodeSync**  08/2025 – 12/2025

- Engineered a **real-time collaborative coding platform** supporting multiple simultaneous users with **sub-second synchronisation, and room-based session management** built on Socket.IO.
- Built **conflict-free live synchronisation** across all connected clients so edits from any user reflect instantly without overwriting concurrent changes from others in the same session.
- Enabled **multi-language code execution across 3 or more languages** with **custom test case input**, keeping the entire coding and testing workflow inside the platform without switching tools mid-session.

**Technology/Tools:** React, TypeScript, Socket.IO, Express.js, CodeMirror.

**AI based HealthMap**  12/2024 – 04/2025

- Built an **AI powered demand forecasting system** for hospital critical supplies, trained on historical consumption data to **predict shortages before they occur** and trigger automated restocking alerts.
- Implemented a model pipeline serialised with joblib that continuously updates on incoming consumption data, with **real-time threshold alerts** surfaced on a React dashboard for hospital staff.
- Owned the complete stack independently across a **React frontend, Node.js and Express API layer, MongoDB data store and a Python ML pipeline**, making this a solo end-to-end production-oriented build.

**Technology/Tools:** ReactJS, Node.js, Express, MongoDB, Python, joblib

## SKILLS

### Languages

Java, JavaScript, TypeScript, Python, Kotlin, Solidity

### Databases & Search Technologies

MySQL, MongoDB, Firebase Realtime Database, SQL, NoSQL, Elasticsearch

### Blockchain

Solidity, Smart Contracts, Decentralised Identity (SSI), Web3

### AI/ML Exposure

Hugging Face Transformers (RoBERTa, T5), Large Language Models (LLMs), Predictive Analytics, Isolation Forest, LightGBM

### Core Computer Science Concepts

Data Structures & Algorithms (DSA), OOP, System Design, Secure Coding Practices, Distributed Systems

### Frameworks & Libraries

ReactJS, Node.js, Express.js, Next.js, Spring Boot (Hibernate), Flutter, Flask

### Tools & Platforms

Git, GitHub, Firebase, Postman, AWS, Streamlit, Docker

### Web Technologies

HTML, CSS, RESTful APIs, Tailwind CSS, WebRTC, Socket.IO